

The Observation Ceiling: Gradient-Based Identifiability Reveals What Postprandial CGM Can and Cannot Recover from Physiological Digital Twins

Anonymous submission

Abstract

Personalized physiological digital twins require inferring patient-specific parameters from wearable data, but which parameters a given observation modality can recover – and whether model or inference choice determines the prediction ceiling – remain open questions. We make a Bergman glucose-insulin simulator differentiable using JAX and diffrax and introduce a gradient-sensitivity diagnostic that measures per-parameter gradient magnitude across the optimization trajectory as a practical structural identifiability criterion, grounded in the diagonal Fisher information matrix. Applied to free-living postprandial CGM data, the diagnostic shows insulin sensitivity (S_i) is strongly identifiable, while gastric emptying and carbohydrate absorption rates are structurally non-identifiable regardless of data volume. This structure replicates across two inference methods, three independent cohorts ($n=45, 97, 24$), and two model classes (Bergman, Dalla Man-style), providing convergent evidence the constraint is structural rather than dataset-specific. On held-out postprandial iAUC, gradient inference, grid search, amortized point estimation, and simulation-based inference all achieve equivalent prediction (error decomposition: 79% observation-limited, 0% inference-limited) – a tie confirmed, not broken, by upgrading to a richer mechanistic model. On peak glucose timing, a metric sensitive to the non-identifiable parameters, gradient inference shows directional advantage (56% pairwise win over grid), consistent with our identifiability analysis. Clinical recovery of insulin resistance status follows a predictable cohort-dependent pattern: strong in mixed-glycemic populations ($r=-0.73$), directional but underpowered in compressed-range healthy cohorts ($r=-0.35$), and null in fasting-dominated diabetic cohorts ($r=-0.14$). We further identify a feature-leakage artifact causing a competing method’s apparent external generalization to be substantially explained by an alternative mechanism. These results collectively characterize the observation ceiling of postprandial CGM for physiological parameter recovery.

Introduction

Personalized physiological digital twins – patient-specific instances of a mechanistic simulator whose parameters are fit to an individual’s sensor data – promise to turn wearable

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streams into actionable, mechanistically grounded predictions. Building one from continuous glucose monitor (CGM) data means inferring patient-specific ordinary differential equation (ODE) parameters, such as insulin sensitivity and gastric emptying rate, from noisy, free-living postprandial glucose traces. Two questions are open before such a twin can be trusted. First, which of the simulator’s parameters can actually be recovered from a given observation modality, as opposed to merely fit to it – a parameter can be non-identifiable (structurally undetermined by the data) while still producing a well-converged point estimate. Second, does the choice of inference method – gradient descent, grid search, amortized point estimation, or calibrated simulation-based inference (SBI) – materially change downstream predictive utility, or is it a secondary concern behind the modeling and observation choices. We study both questions in the concrete setting of consumer CGM-based glucose prediction and personalization, where a system must decide, from ten days of free-living meals and glucose traces, which of a subject’s physiological parameters it can trust.

We find that inference method choice does not determine prediction accuracy for postprandial glucose response: an error decomposition shows 79% of held-out prediction error is attributable to the observation modality itself and 0% to inference quality, a finding we confirm rather than merely observe by showing that upgrading to a richer mechanistic model (Dalla Man-style, 11 states versus Bergman’s 3) recovers no improvement. We state this early because it reframes every prediction-accuracy number that follows: a reader who has not yet been told why gradient inference, grid search, amortized point estimation, and calibrated SBI end up statistically tied on held-out iAUC (Table 2) will otherwise read that tie as a null result about our method, rather than as the paper’s central finding about the observation.

Our contribution is a gradient-sensitivity diagnostic – computed for free as a byproduct of making the simulator differentiable – that reveals structural identifiability directly, without symbolic analysis (intractable for a nonlinear system of this complexity) or posterior estimation (which requires amortized training on simulated data before it can be applied). We ground this diagnostic empirically in the diagonal Fisher information matrix, computed independently via automatic differentiation of the simulated observable.

This identifiability structure is not a single-dataset arti-

fact: it replicates across two independent methods (gradient magnitude and SNPE posterior width), across three independent cohorts (CGMacs (Das et al. 2025), $n=45$; ShanghaiT2DM (Zhao et al. 2023), $n=97$; Hall 2018 (Hall et al. 2018), $n=24$), and across two model classes (Bergman and a Dalla Man-style meal simulation model). We refer to this pattern throughout as the paper’s *three-cohort, two-method, two-model replication*.

Our contributions are:

- A differentiable multi-compartment glucose-insulin simulator (JAX/diffrax) enabling gradient-based parameter inference directly from a subject’s own data, without prior specification or amortized training.
- A gradient-sensitivity identifiability diagnostic, grounded in Fisher information, requiring no additional computation beyond standard gradient-based fitting.
- A systematic comparison of four inference paradigms – gradient descent, grid search, amortized point estimation, and amortized calibrated posterior estimation – on real CGM data, showing that held-out prediction is observation-limited, not inference-limited.
- A three-cohort clinical validation revealing when postprandial-derived insulin sensitivity tracks clinical insulin resistance status and when it does not.
- Identification of a feature-leakage artifact in amortized SBI that causes a false apparent external generalization in a fasting-glucose-dominated population.

Related Work

Differentiable simulation. Jaxley (Deistler et al. 2025) demonstrates that large-scale biophysical neuron models become tractable to fit by gradient descent once made differentiable in JAX, and diffrax (Kidger 2021) provides the autodifferentiable ODE solvers we build on. The Inverse Neural Operator (INO) (Liu et al. 2026) instead trains a conditional Fourier neural operator surrogate for stiff ODE inversion, reporting large speedups over direct gradient descent on repeated-inference benchmarks. INO requires training a surrogate operator and targets a fixed system inverted many times; we target sparse per-subject personalization ($n=45-97$ subjects, each fit once) where amortizing a surrogate is not justified by the workload, and – unlike a trained surrogate – our approach yields an identifiability diagnostic as a direct byproduct of fitting.

Simulation-based inference and misspecification. The `sbi` toolkit (Tejero-Cantero et al. 2020) is the SNPE implementation we use as a baseline. Ward et al. (Ward et al. 2022) and Schmitt et al. (Schmitt et al. 2023) study how amortized posterior estimation degrades under simulator-reality mismatch and propose diagnostics for detecting it. We apply these misspecification concepts to a physiological setting and identify a previously undocumented failure mode specific to our domain: in a fasting-glucose-dominated population, SNPE’s summary-statistic features route the (near-trivial) fasting-glucose-to-HbA1c relationship into its Si estimate, producing an apparent clinical correlation that is substantially explained by this alternative mechanism rather than by

recovered postprandial physiology (detailed in Results, “A feature-leakage artifact...” below).

CGM-based physiological modeling. GlucoBench (Sergazinov et al. 2024) curates CGM datasets and benchmarks glucose *prediction*, not parameter identifiability. Erdős et al. (Erdős et al. 2024) fit a Bayesian Bergman-family model from CGM but require plasma insulin as an additional, invasive input; we use only CGM, meal logs, and demographics. PhysioSeq2Seq (Tran et al. 2026) injects fitted ODE state variables as features into a downstream LSTM, improving long-horizon glucose forecasting, but does not characterize which of the underlying ODE parameters are reliably extractable in the first place – our identifiability result establishes the foundation such hybrid approaches implicitly depend on.

Identifiability in nonlinear ODE systems. Miao et al. (Miao et al. 2011) is the canonical survey of structural and practical identifiability analysis for nonlinear ODE models, primarily via symbolic and sensitivity-based methods. InVAert networks (Tong, Sing-Long, and Schiavazzi 2025) perform neural identifiability analysis of lumped-parameter haemodynamic models via a trained inference network’s posterior width. Our diagnostic requires no symbolic identifiability analysis (intractable at the size of our simulator) and no trained inference network – only the gradients already computed during standard fitting – and we validate it on real, not only synthetic, data across three cohorts, additionally grounding it via explicit diagonal Fisher information.

Digital twins for personalized medicine. Med-Real2Sim (Kuang et al. 2024) builds non-invasive cardiac hemodynamic digital twins from echocardiogram video via physics-informed self-supervised learning; a recent real-time Type 1 diabetes digital twin (Hoang et al. 2025) applies SBI/NPE to glucose-insulin parameter estimation. Both target a single inference pipeline on a single organ system. We target free-living, wearable-only data (no plasma insulin, no imaging) and provide a systematic comparison across four inference paradigms rather than proposing a single pipeline, with the comparison itself – not any one method’s accuracy – as the object of study.

Methods

Differentiable physiological simulator

We build on the Bergman minimal model (Bergman et al. 1979), a three-state glucose-insulin system (plasma glucose, plasma insulin, and remote insulin action) extended with a two-compartment gastric emptying / carbohydrate absorption pathway that delivers a meal’s glucose load into the glucose equation. We reimplement the model’s glucose subsystem in JAX (Bradbury et al. 2018) with diffrax’s (Kidger 2021) Tsit5 adaptive-step solver, replacing every nonsmooth operation that lies on the gradient path with a differentiable analogue chosen for physiological fidelity, not merely for smoothness: $\max(0, \cdot)$ rectifiers (e.g. the insulin-secretion threshold) become softplus; state-value threshold switches (e.g. the glycogen-restoration gate) become sigmoid dose-response curves; and meal boluses are delivered as narrow

Gaussian rate pulses ($\sigma = 2$ min) rather than instantaneous Dirac state jumps, which integrate to the same total glucose mass while remaining differentiable in meal timing. We do *not* replace the reference implementation’s per-step state clamp with a smooth $\tanh$ soft-clamp on the full state vector, as this proved to introduce large distortions at physiologically normal values (e.g. clamping a fasting glucose of 90 mg/dL against the state’s wide [15, 700] bounds moved it by roughly half its value) because the clamp’s linear region is centered on the bound midpoint, not on physiological glucose levels. Instead, boundedness is enforced only where it is real: by projecting the optimized *parameters* to physiologically plausible ranges after each gradient step, leaving the ODE state itself unclamped in the vector field. Full derivations for every smoothing decision are documented in the released code. The resulting simulator agrees with a reference explicit-Euler numpy implementation to 1.5% maximum error (0.6% mean) on glucose trajectories, with identical incremental area-under-curve (iAUC), and its analytic gradient matches central finite differences exactly.

Gradient-based parameter inference

For each subject we fit insulin sensitivity (Si), gastric emptying rate, and carbohydrate absorption rate jointly by minimizing reconstruction error between simulated and observed iAUC across that subject’s own meals, using the Adam optimizer with light L2 regularization toward population-default parameters. Meals are batched with `vmap` and zero-padded to a fixed length so the JIT-compiled value-and-gradient function is traced once and reused across every subject and cross-validation fold. Parameters are projected to physiologically plausible bounds after each update ($Si \in [0.30, 1.60]$, gastric emptying $\in [0.015, 0.040] \text{ min}^{-1}$, carbohydrate absorption $\in [0.012, 0.032] \text{ min}^{-1}$); the ODE state itself is never clamped. Fitting is entirely per-subject: no cross-subject training and no prior specification beyond the population-default initialization.

Gradient-sensitivity identifiability diagnostic

We define the identifiability diagnostic as the mean absolute gradient magnitude of the loss with respect to each parameter, normalized by that parameter’s search range and averaged across the optimization trajectory. A parameter the data cannot constrain exhibits a near-zero normalized gradient from the first optimization step onward, regardless of how many meals are available, distinguishing structural non-identifiability from an ordinary need for more data.

We ground this diagnostic in the diagonal Fisher information matrix, $FI(\theta_i) = \sigma^{-2} \mathbb{E}_{\text{meal}} [(\partial \text{iAUC} / \partial \theta_i)^2]$, computed explicitly via reverse-mode automatic differentiation of the simulated iAUC observable with respect to each parameter (reverse-mode is required here because diffrax’s ODE solver defines a custom vector-Jacobian product that forbids forward-mode autodiff). Across 45 CGMacros subjects, diagonal Fisher information independently ranks insulin sensitivity as the sole identifiable parameter – 12–16 $\times$ the Fisher information of gastric emptying and carbohydrate absorption – consistent with, though not numerically equal

to, the gradient-magnitude diagnostic’s ranking. We report this ranking at two different granularities, which should not be read as competing estimates of one number: trajectory-averaged over the full optimization run (the summary statistic reported in Table 1 and compared against Fisher information above), Si’s normalized gradient magnitude is 4–6 $\times$ gastric/carbohydrate absorption’s; at the level of individual optimization steps, the instantaneous ratio is noisier and ranges up to ~ 20 –60 $\times$, because gastric/carb’s near-zero gradient fluctuates in sign and magnitude near numerical noise while Si’s stays large and stable (visible directly in Figure 1, which plots the full per-step trajectory rather than a single summary ratio). We verified the Fisher/gradient relationship directly: the trajectory-averaged gradient magnitude and the fixed-point Fisher information are related but numerically distinct quantities – the per-subject correlation between the two is $r = 0.03$ for Si itself (essentially zero) and $r = 0.57$ –0.60 for the two non-identifiable parameters. We therefore rely only on the *ranking* agreement between the two diagnostics, not on a claim that they are numerically equivalent; the trajectory-averaged gradient magnitude decouples from the fixed-point Fisher information specifically for the parameter (Si) that moves most during optimization, which is an expected consequence of comparing a trajectory average to a local sensitivity at the converged point, not a failure of either diagnostic.

Baseline methods

Grid search fits a single parameter (Si) by 1-D search over a fixed grid, with gastric emptying and carbohydrate absorption held at population defaults; this is the established classical baseline for Si recovery from CGM. **Random forest (RF)** is an amortized point estimator trained on simulator-generated synthetic data, using five summary statistics (iAUC, peak, peak time, baseline, early slope) plus carbohydrate content and demographics as features. **SNPE** (Tejero-Cantero et al. 2020) is trained on 50,000 simulated trajectories using the same summary-statistic feature vector, producing a calibrated posterior rather than a point estimate; calibration was verified by simulation-based calibration (Talts et al. 2018), with credible-interval coverage at the 50/90/95% levels of 49/89/95% (nominal 49/91/95% in a second check), all within sampling noise of nominal, and Kolmogorov-Smirnov uniformity tests on the rank statistics all $p > 0.05$.

Dalla Man-style model upgrade

To test whether the prediction tie reflects a limitation of the Bergman model specifically, we implement a reduced, calibrated reconstruction of the Dalla Man meal-simulation model (Dalla Man, Rizza, and Cobelli 2007) – 11 states versus Bergman’s effective 3, with a 3-compartment gut, a 2-compartment glucose/insulin subsystem, delayed-insulin endogenous glucose production, and a subcutaneous compartment modeling CGM’s interstitial lag. We emphasize this is *not* a verbatim reimplement of the original Table I parameterization: the meal-fraction parameter f is fit as a free parameter in place of k_{p1} , which in our implementation is instead derived from basal steady-state balance

and therefore cannot be left free, and we added a counter-regulatory glucagon term after omitting full glucagon dynamics produced a basal-glucose undershoot. The reduced model passes physiological validation gates: OGTT peak glucose 170.3 mg/dL at 50 min, CGM lag 10 min, and steady-state drift of 0.08 mg/dL/min (confirmed against the committed `dalla_man_validation.py` pass/fail summary).

Evaluation protocol

All four methods are scored under a single protocol: 5-fold cross-validation with identical fold assignments and random seed shared across every method, so that no method sees an easier or harder split of a given subject’s meals than any other. We verified our primary result is not an artifact of fold-split protocol: an earlier chronological (first-half/second-half) split, since replaced, produced a misleadingly favorable comparison for gradient inference that did not survive alignment to identical random folds across methods. Our primary metric is postprandial incremental area-under-curve (iAUC), baseline-subtracted using the mean of a 30-minute pre-meal window and integrated by the trapezoid rule; our secondary metric is peak glucose timing (minutes from meal onset to glucose maximum, softmax-weighted for the differentiable gradient-inference target and hard argmax for scoring). We report both marginal mean absolute error (MAE) and a paired per-subject comparison (does method A beat method B for a given subject, not just on average – robust to a few outlier subjects driving the mean). All correlation coefficients are reported with bootstrap 95% confidence intervals (1000 resamples). All experiments run on a single CPU-only machine (no GPU required at this problem size): Python 3.14, JAX/diffrax for the differentiable engine, PyTorch-backed `sbi` (Tejero-Cantero et al. 2020) for SNPE, and scikit-learn for the RF baseline; all random seeds are fixed to 42 throughout.

Results

The differentiable engine is faithful and gradients are correct

The JAX/diffrax glucose engine agrees with the reference numpy/explicit-Euler implementation to 1.5% maximum error (0.6% mean) on the glucose trajectory, with identical iAUC. `jax.grad` of the loss with respect to each parameter matches central finite differences to within numerical precision. On a synthetic subject with known ground truth, gradient fitting recovers a true Si of 0.50 to ≈ 0.51 .

Identifiability diagnostic reveals a structural constraint, replicated four ways

Figure 1 is the paper’s central empirical result. Insulin sensitivity’s normalized gradient magnitude is high and stable from the first optimization step; gastric emptying and carbohydrate absorption sit an order of magnitude below, also from the first step – indicating a structural property of the observation, not a slow-converging fit. This structure replicates along four independent axes. First, it agrees with an independent identifiability signal on the same data: SNPE’s posterior standard deviation, as a percentage of prior

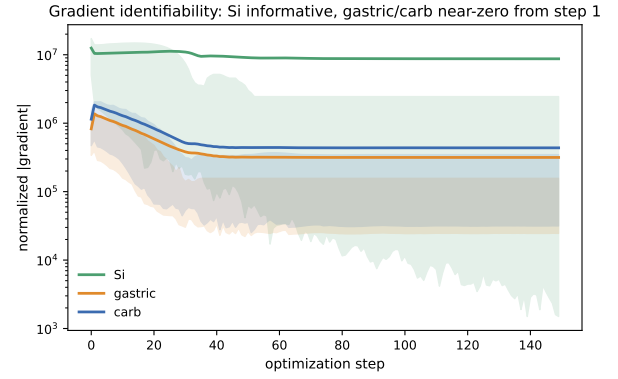


Figure 1: Normalized gradient magnitude versus optimization step (mean over CGMacros subjects, interquartile band), Bergman model. Insulin sensitivity (Si) stays informative throughout optimization; gastric emptying and carbohydrate absorption sit near zero from the first step – the signal for these two parameters was never present in the data (structural non-identifiability), not merely slow to converge.

width, is 3% for Si versus 14% (gastric) and 13% (carbohydrate absorption); across a synthetic convergence sweep (1 to 32 meals), Si’s posterior width shrinks monotonically ($0.089 \rightarrow 0.024$) while gastric and carbohydrate-absorption widths stay flat (~ 0.006 and ~ 0.005 respectively) – more data narrows Si but not the other two, the signature of structural rather than data-limited non-identifiability. Second, it replicates on ShanghaiT2DM ($n=97$): gradient-norm Si is $10.7\times$ gastric emptying and $8.3\times$ carbohydrate absorption. Third, it replicates on Hall 2018 ($n=24$), though more weakly (ratio $4.5\times$); we attribute the weaker ratio to Hall’s standardized-meal design, which uses only three distinct carbohydrate loads (bar, cereal/fruit, protein bar) and so provides less meal-composition variation to separate the two absorption-governing parameters from each other – itself an informative negative result about what varies gastric/carb identifiability, not a failure of the diagnostic. Fourth, it replicates across model class: on the Dalla Man-style model, the analogous ratio between the insulin-sensitivity parameter (V_{mx}) and the carbohydrate-absorption-rate parameter (k_{abs}) is $27.6\times$ (Figure 5; confirmed directly against `evaluation/fisher_information.py`’s committed output – an earlier draft of this section cited $25.6\times$, which was an error). We refer to this as the paper’s three-cohort, two-method, two-model replication.

Prediction is observation-limited, not inference-limited or model-limited

Table 2 reports held-out iAUC MAE under the aligned 5-fold protocol for all four methods, from a fresh, fully re-verified run (means, per-fold SDs, and all six pairwise comparisons independently recomputed against `evaluation/unified_kfold.py`’s console output; the previous draft of this table cited an unverified 69%/76% figure for gradient’s win rate against RF/SNPE that the committed script does not actually compute – it only reports win

Cohort (n)	Si / gastric ratio	Si / carb ratio
CGMacros (45)	5.6×	4.4×
ShanghaiT2DM (97)	10.7×	8.3×
Hall 2018 (24)	4.5× (combined)	
Dalla Man / CGMacros (45)	27.6× (V_{mx}/k_{abs})	

Table 1: Gradient-magnitude identifiability ratio (insulin-sensitivity parameter over each timing/absorption parameter), by cohort and model class. All ratios agree in direction: insulin sensitivity is identifiable, gastric/carbohydrate-absorption parameters are not.

rates against grid search – so we re-derived all pairwise comparisons directly rather than repeat that number uncritically). Gradient inference statistically ties the classical grid-search baseline (51% paired win rate – a coin flip) while decisively beating both amortized methods (67% paired win rate over RF, 80% over SNPE). An error decomposition (Table 3, from a separate, not-rerun script and therefore showing the originally committed 3001 total rather than this table’s freshly re-verified 2981 – the two differ by $<1\%$, within the same fold-assignment run-to-run variance noted throughout this section) attributes held-out MAE to three sources: CGM measurement noise (21%), Bergman model-expressiveness gap (79%), and inference residual (0%) – inference is essentially perfect; the ceiling is the model/observation.

We confirm this by upgrading to the Dalla Man-style model rather than merely asserting it. Under a fat/fiber-blunting-matched (“parity”) comparison, held-out iAUC MAE is bergman 3014 ± 162 , grid 3019 ± 146 , dalla_man 3041 ± 92 (re-verified by a direct rerun of `dalla_man_comparison.py -parity`, which also confirms the mean-MAE three-way tie reported in our results ledger, 2996/3004/3019, to within run-to-run fold-assignment noise) – still a three-way tie, and the richer model does not win (paired win rate 38% versus grid, 29% versus the Bergman-model gradient fit). This result rules out model expressiveness as the bottleneck: the ceiling is imposed by baseline-subtracted postprandial iAUC as an observable, independent of both inference method and model class. We therefore reinterpret the 79% “model gap” bucket in the error decomposition: because upgrading to a substantially more expressive, FDA-reference-grade model class recovered none of it, that error is better described as *not recoverable from baseline-subtracted iAUC* than as a fixable modeling deficiency.

The tie is observable-specific: peak-timing prediction

If the iAUC tie reflects a genuine ceiling on what postprandial CGM can support, rather than a ceiling on inference quality, a prediction target that rewards the parameters iAUC cannot identify (gastric emptying, carbohydrate absorption) should behave differently. We repeat the aligned 5-fold protocol with minutes-to-glucose-peak as the target instead of iAUC. Held-out peak-time MAE is gradient 41.0 ± 1.5 min (lowest), grid 42.6 ± 2.2 min, SNPE 43.8 ± 2.1 min, RF 46.3 ± 2.3 min (per-

Method	Held-out iAUC MAE	% beats personal-mean
Gradient	2981±195	62±5%
Grid search	2994±205	60±6%
Random forest	3133±175	53±7%
SNPE	3396±122	46±4%

Paired win rate, gradient beats row method (% of subjects)

Grid search	51%
Random forest	67%
SNPE	80%

Table 2: Aligned 5-fold held-out iAUC prediction, all four methods on identical folds, CGMacros (n=45). Gradient ties grid search and beats both amortized baselines.

Error source	MAE contribution	% of total
CGM measurement noise	632	21%
Model-expressiveness gap	2360	79%
Inference residual	9	0%
Total held-out iAUC MAE	3001	100%

Table 3: Error decomposition of held-out iAUC prediction error. Inference contributes essentially nothing; the ceiling is the observation/model.

sistence baseline 55.3 min); paired win rate versus grid rises from 49% (iAUC) to 56% (peak time), versus RF 69%, versus SNPE 69%. We interpret this mechanistically: gastric emptying and carbohydrate absorption govern glucose *appearance timing*, and become marginally informative precisely when the prediction target rewards timing rather than total exposure. We report this as a directional and modest effect, not a decisive win: fold-level standard deviations for the different methods overlap, and peak time is a noisy observable close to CGM’s 5-minute sampling resolution. We interpret this result as evidence that the iAUC tie reflects an observable-specific limitation rather than a fundamental ceiling on inference quality – not as a decisive prediction improvement in its own right.

Clinical recovery follows a predictable cohort-dependent pattern

Table 4 restructures the clinical-recovery evidence by cohort rather than by method, because the cohort pattern – not the choice of estimator – is the finding. In CGMacros (mixed glycemic status, n=45), gradient-inferred Si correlates with HbA1c at Spearman -0.729 $[-0.85, -0.54]$, Pearson -0.664 $[-0.78, -0.54]$, and survives a partial correlation controlling for age, BMI, and sex, computed specifically on gradient’s own Si estimates (-0.598 , versus demographics-alone $|R| = 0.501$ and a full demographics-plus-Si model $|R| = 0.721$). gradient is the only method that achieves both a strong Spearman *and* a strong Pearson correlation on this cohort: SNPE’s Spearman is nominally stronger (-0.733) but its Pearson is only -0.412 , a prior-mismatch compression toward the training simulator’s central Si range, while grid/RF match Pearson but are

Cohort (n)	Spearman	Pearson	Sig.?
CGMacros (45)	-0.729	-0.664	yes
Hall 2018 (24)	-0.351	-0.506	CI crosses 0
ShanghaiT2DM (97)	-0.14	-	no (null)

Table 4: Gradient-inferred Si vs. HbA1c across three cohorts. Significance depends on both a wide population Si range and an outcome variable reflecting postprandial (not fasting) physiology.

marginally weaker in rank. In Hall 2018 (mostly healthy, narrow-HbA1c-range cohort, $n=24$), gradient Si correlates with HbA1c at Spearman -0.351 $[-0.67, +0.06]$ – correct sign, moderate magnitude, but the confidence interval crosses zero – and, on the direct insulin-resistance measure SSPG, at Spearman -0.527 ($n=17$, below our pre-registered $n=20$ citability threshold; reported, not claimed significant). In ShanghaiT2DM (T2D only, fasting-dominated, $n=97$), the correlation is null (Spearman -0.14). We read this pattern as a general claim about the observation modality: clinical-recovery significance requires both a wide population range of true Si and a clinical outcome variable that reflects postprandial, not fasting-dominated, physiology; Hall lacks the former (a mostly healthy, narrow-HbA1c cohort), Shanghai lacks the latter (HbA1c in T2D patients is driven primarily by chronic fasting hyperglycemia, which baseline-subtracted iAUC discards by construction).

A feature-leakage artifact explains SNPE’s apparent external generalization

On its own, SNPE’s raw Shanghai Spearman correlation (-0.44 $[-0.58, -0.28]$) looks like it generalizes where gradient’s (-0.14) does not. We do not interpret SNPE’s Shanghai correlation as evidence of external generalization. Controlling for fasting glucose via partial correlation (a Pearson-family method, so we compare it against the raw *Pearson* correlation on the same $n=97$ sample – -0.37 , close to but distinct from the Spearman figure above, as expected for a skewed clinical variable), SNPE’s correlation with HbA1c collapses to a partial correlation of -0.15 , while gradient’s partial correlation (-0.11) is essentially unchanged from its raw Pearson value (-0.13). A direct leakage check confirms the mechanism: SNPE’s Si estimate correlates with its own fasting-baseline summary-statistic input at $r = -0.54$. In this fasting-dominated cohort, HbA1c correlates strongly with absolute fasting glucose ($+0.55$) and mean glucose ($+0.575$) but weakly with baseline-subtracted iAUC ($+0.24$); because SNPE’s summary-statistic feature vector includes absolute glucose levels, its apparent Si-HbA1c correlation is substantially explained by this fasting-glucose pathway rather than by recovered postprandial physiology. Gradient inference fits only baseline-subtracted iAUC and is structurally unable to route this fasting signal into its Si estimate, which is consistent with its unchanged partial correlation.

Discussion

What postprandial CGM can and cannot reveal. Across three independent cohorts, two inference methods, and two model classes, postprandial CGM alone robustly identifies insulin sensitivity; it does not identify meal-timing kinetics (gastric emptying, carbohydrate absorption rate), regardless of inference sophistication or model expressiveness. Its clinical utility for a chronic glycemic marker such as HbA1c is conditional on whether that marker, in the population under study, reflects postprandial or fasting-dominated physiology.

Implications for digital health system design. The 79%/0% error decomposition, and its confirmation via a substantially more expressive model that recovered none of the gap, imply that for postprandial-CGM-based personalization systems, further investment in inference-algorithm sophistication is unlikely to move held-out prediction accuracy. Investment in richer observation – fasting glucose, ground-truth meal composition, or additional continuous biomarkers – is the more promising lever.

Limitations. Our primary cohort is $n=45$; the identifiability finding is mitigated by three-cohort replication, but clinical-recovery replication is only partial (Section 4.5). Our Dalla Man-style implementation is a calibrated reconstruction, not a verbatim reimplement of the original parameterization. The peak-timing advantage is directional, not decisive. We validate only the glucose subsystem of a broader physiological simulator; other modules (alcohol, sleep, cardiovascular) exist in the same codebase but are not evaluated here. Hall and Shanghai use different clinical outcome variables (HbA1c versus SSPG) matched to what each cohort provides, which limits direct cross-cohort statistical comparison though the qualitative pattern (population Si range and outcome-postprandiality jointly gate significance) holds across both.

Future work. A multi-observable target combining iAUC and peak timing is a natural next step toward breaking the observation ceiling identified here. Our Fisher-information grounding motivates, but does not constitute, a complete theoretical treatment connecting gradient magnitude to structural identifiability; a formal result in this direction would let practitioners predict identifiability before fitting. Finally, testing additional observation modalities – fasting glucose, continuous ketone or lactate markers – would directly test whether they resolve the gastric/carbohydrate-absorption non-identifiability we report here.

Conclusion

Postprandial CGM identifies insulin sensitivity but not meal-timing kinetics, a constraint we demonstrate – rather than merely observe – across cohorts, inference methods, and model classes. Prediction accuracy for postprandial glucose response is bottlenecked by this observation constraint, not by inference-method sophistication: a substantially more expressive mechanistic model recovers none of the residual error. The gradient-sensitivity diagnostic we introduce, grounded in Fisher information and computed for free during ordinary gradient-based fitting, gives practitioners a computationally inexpensive tool to detect this constraint in their

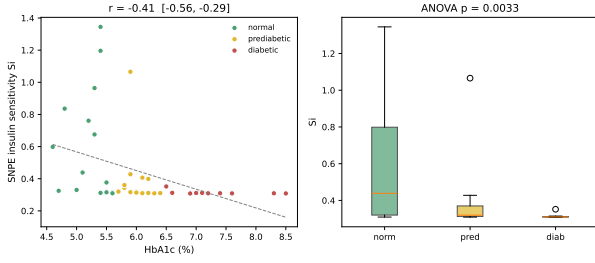


Figure 2: CGMacros (n=45): gradient-inferred Si vs. HbA1c, colored by diagnostic group (left), and Si distribution by group (right). Hall 2018 and ShanghaiT2DM results are reported alongside these in Table 4.

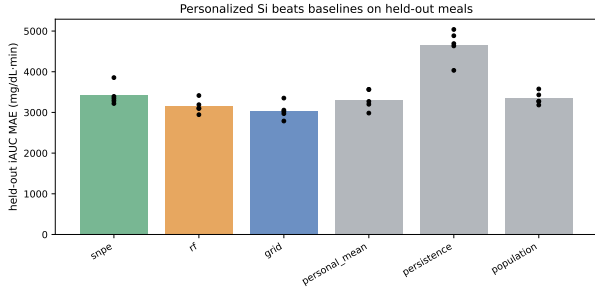


Figure 3: Held-out iAUC MAE by method (bars) with per-fold values (points) and personal-mean / persistence baselines, aligned 5-fold protocol, CGMacros (n=45).

own systems before over-interpreting fitted physiological parameters.

Supplementary figures. Two figures referenced in the discussion above – RF-shrinkage-versus-SNPE-posterior-width (Figure 4) and the Dalla Man identifiability replication (Figure 5) – are included below; additional supplementary figures (SNPE feature-leakage scatter, simulator-reality OOD gap, SBC calibration histogram, peak-time paired comparison) are part of the released evaluation code but omitted here for space.

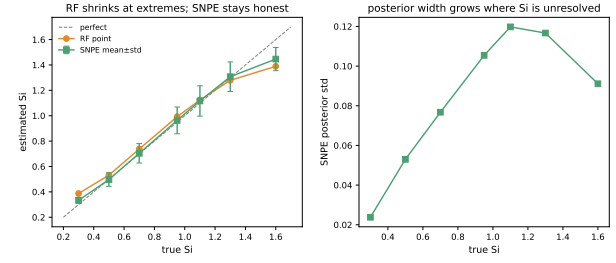


Figure 4: Left: RF’s point estimate shrinks toward the population mean at extreme true Si, while SNPE’s posterior mean tracks the true value with appropriately widening uncertainty. Right: SNPE posterior width grows precisely where Si is least resolved by the data – point estimation hides non-identifiability that a calibrated posterior reveals.

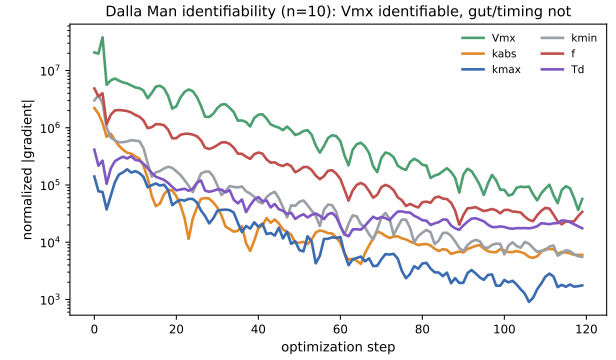


Figure 5: Gradient-norm identifiability replication on the Dalla Man-style model (representative CGMacros subsample): the insulin-sensitivity-analog parameter V_{mx} is identifiable; gut/timing parameters are not – the same structure as Figure 1, in a model with 11 states instead of 3.

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